

演

問1

$$X_1, \dots, X_n, \text{ i.i.d. } \sim N(\mu, \sigma^2)$$

$\bar{X}, S^2$ は 標本平均, 標本分散

$\bar{X} + \sqrt{n} S$ の 平均, 分散

$$V^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\frac{(n-1)V^2}{\sigma^2} \sim \chi_{n-1}^2$$

$$\frac{(n-1)V^2}{\sigma^2} = \frac{nS^2}{\sigma^2}$$

$$Y = \frac{nS^2}{\sigma^2} \quad S^2 = \frac{\sigma^2 Y}{n}$$

$$S = \frac{\sigma}{\sqrt{n}} \sqrt{Y}$$

$$E[S] = E\left[\frac{\sigma}{\sqrt{n}} \sqrt{Y}\right]$$

$$= \frac{\sigma}{\sqrt{n}} E[\sqrt{Y}]$$

$$E[\sqrt{Y}] = \int_0^\infty \sqrt{y} \frac{1}{\Gamma(\frac{n-1}{2})} \left(\frac{1}{2}\right)^{\frac{n-1}{2}} \cdot y^{\frac{n-1}{2}-1} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= \int_0^\infty \frac{1}{\Gamma(\frac{n-1}{2})} \left(\frac{1}{2}\right)^{\frac{n-1}{2}} \cdot y^{\frac{n-1}{2}-1} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= \frac{\sqrt{2} \Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})} \int_0^\infty \frac{1}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} \cdot y^{\frac{n}{2}-1} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= \sqrt{2} \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})}$$

$$E[\bar{X} + \sqrt{n} S]$$

$$= E[\bar{X}] + \sqrt{n} E[S]$$

$$= E\left[\frac{1}{n} \sum_{i=1}^n X_i\right] + \sqrt{n} \cdot \frac{\sigma}{\sqrt{n}} \cdot \sqrt{2} \cdot \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})}$$

$$= \mu + \sqrt{2} \sigma \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})}$$

$$E[(\bar{X} + \sqrt{n} S)^2]$$

$$= E[\bar{X}^2 + \sqrt{n} \bar{X} S + n S^2]$$

$$= E[\bar{X}^2] + \sqrt{n} E[\bar{X} S] + n E[S^2]$$

$$V_{\text{var}}(\bar{X}) = V_{\text{var}}\left[\frac{1}{n} \sum_{i=1}^n X_i\right]$$

$$= \frac{1}{n^2} (V_{\text{var}}(X_1) + \dots + V_{\text{var}}(X_n))$$

$$= \frac{\sigma^2}{n}$$

$$E[\bar{X}^2] = V_{\text{var}}(\bar{X}) + E[\bar{X}]^2$$

$$= \frac{\sigma^2}{n} + \mu^2$$

$$E[\bar{X} S] = E[\bar{X}] \cdot E[S]$$

$$= \mu \cdot \frac{\sigma}{\sqrt{n}} \cdot \sqrt{2} \cdot \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})}$$

$$E[S^2] = E\left[\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2\right]$$

$$= \sigma^2 - \frac{\sigma^2}{n} = \frac{1}{n} (n-1) \sigma^2$$

$$V_{\text{var}}(\bar{X} + \sqrt{n} S)$$

$$= \frac{\sigma^2}{n} + \mu^2 + \sqrt{2} \mu \sigma \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})} + (n-1) \sigma^2$$

$$- \mu^2 - \sqrt{2} \mu \sigma \frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})} + 2 \sigma^2 \left(\frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})}\right)^2$$

$$= (n-1 + \frac{1}{n} + 2 \left(\frac{\Gamma(\frac{n}{2})}{\Gamma(\frac{n-1}{2})}\right)^2) \sigma^2$$

Fr 2

$$Z \sim N(0,1)$$

$$Y \sim \chi_m^2$$

$$T = \frac{Z}{\sqrt{\frac{Y}{m}}}$$

$$f_T(t) = f_{T|Y}(T|Y=Y) f_Y(Y)$$

$$T|Y=Y \sim N(0, \frac{m}{Y})$$

$$Y \sim \chi_m^2$$

$$\frac{2}{\sqrt{\frac{Y}{m}}}$$

$$P(T \leq t)$$

$$= \int_{-\infty}^t \int_0^{\infty} f_{T|Y}(t|Y=Y) f_Y(Y) dY dt$$

$$= \int_0^{\infty} \left(\int_{-\infty}^t f_{T|Y}(t|Y=Y) dt \right) f_Y(Y) dY$$

$$= \int_0^{\infty} F_{T|Y}(t|Y) \cdot f_Y(Y) dY$$

$$= \int_0^{\infty} P\left(\frac{Z}{\sqrt{\frac{Y}{m}}} \leq t\right) \cdot f_Y(Y) dY$$

$$= \int_0^{\infty} P(Z \leq \sqrt{\frac{Y}{m}} t) f_Y(Y) dY$$

$$f_T(t) = \int_0^{\infty} f_{T|Y}(t|Y) \cdot f_Y(Y) dY$$

$$= \int_0^{\infty} \frac{1}{\sqrt{2\pi \frac{m}{Y}}} \exp\left\{-\frac{t^2}{2 \frac{m}{Y}}\right\} \cdot \frac{1}{\Gamma(\frac{m}{2})} \left(\frac{1}{2}\right)^{\frac{m}{2}} Y^{\frac{m}{2}-1} \exp\left\{-\frac{Y}{2}\right\} dY$$

$$= \frac{1}{\sqrt{2\pi m}} \cdot \frac{\left(\frac{1}{2}\right)^{\frac{m}{2}}}{\Gamma(\frac{m}{2})} \int_0^{\infty} Y^{\frac{m-1}{2}} \exp\left\{-\frac{(m+t^2)Y}{2m}\right\} dY$$

$$= \frac{1}{\sqrt{2\pi m}} \cdot \frac{\left(\frac{1}{2}\right)^{\frac{m}{2}}}{\Gamma(\frac{m}{2})} \int_0^{\infty} Y^{\frac{m-1}{2}} \exp\left\{-\frac{Y}{\frac{2m}{m+t^2}}\right\} dY$$

$$U = \frac{2m}{m+t^2}$$

$$= \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m}{2})} \frac{\left(\frac{1}{2}\right)^{\frac{m}{2}}}{\sqrt{2\pi m}} \int_0^{\infty} \frac{1}{\Gamma(\frac{m+1}{2})} \frac{1}{U} \left(\frac{Y}{U}\right)^{\frac{m+1}{2}-1} \exp\left\{-\frac{Y}{U}\right\} dY$$

$$= \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m}{2})} \cdot \frac{1}{\sqrt{2\pi m}} \cdot \left(\frac{1}{2}\right)^{\frac{m}{2}} \cdot \left(\frac{2m}{m+t^2}\right)^{\frac{m+1}{2}}$$

$$= \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m}{2})} \cdot \frac{1}{\sqrt{\pi m}} \left(\frac{m}{m+t^2}\right)^{\frac{m+1}{2}}$$

問 3

$$E[T], \text{var}(T)$$

$$E[T] = E[E[T|Y=r]]$$

$$= E[0] = 0$$

$$\text{var}(T) = E[\text{var}(T|Y)] + \text{var}(E[T|Y])$$

$$= E\left[\frac{m}{Y}\right] + 0$$

$$= E\left[\frac{m}{Y}\right]$$

$$= m E\left[\frac{1}{Y}\right]$$

$$= m \int_0^{\infty} y^{-1} \cdot \frac{1}{\Gamma(\frac{m}{2})} \left(\frac{1}{2}\right)^{\frac{m}{2}} y^{\frac{m}{2}-1} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= m \int_0^{\infty} \frac{1}{\Gamma(\frac{m}{2})} \left(\frac{1}{2}\right)^{\frac{m}{2}} y^{\frac{m}{2}-1} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= \frac{m \Gamma(\frac{m}{2}-1)}{2 \Gamma(\frac{m}{2})} \int_0^{\infty} \frac{1}{\Gamma(\frac{m-1}{2})} \cdot \left(\frac{1}{2}\right)^{\frac{m-1}{2}} y^{\frac{m-1}{2}-1} \exp\left\{-\frac{y}{2}\right\} dy$$

$$= \frac{m}{2(\frac{m}{2}-1)} = \frac{m}{m-2}$$

$$E[T^2]$$

$$= \int_{-\infty}^{\infty} t^2 \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m}{2})} \frac{1}{\sqrt{m\pi}} \frac{1}{\left(1+\frac{t^2}{m}\right)^{\frac{m+1}{2}}} dt$$

$t = 2R+1$ のとき 奇数だから 0

$t = 2R$ のとき

$$\int_{-\infty}^{\infty} t^{2R} \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m}{2})} \frac{1}{\sqrt{m\pi}} \left(1+\frac{t^2}{m}\right)^{-\frac{m+1}{2}} dt$$

$$= 2 \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m}{2})} \cdot \frac{1}{\sqrt{m\pi}} \int_0^{\infty} t^{2R} \left(1+\frac{t^2}{m}\right)^{-\frac{m+1}{2}} dt$$

$$u = \left(1+\frac{t^2}{m}\right)^{-1} \quad \begin{array}{c|cc} t & 0 & \rightarrow \infty \\ \hline u & 1 & \rightarrow 0 \end{array}$$

$$\begin{aligned} 1+\frac{t^2}{m} &= \frac{1}{u} & t^2 &= \frac{m}{u} - m & \frac{du}{dt} &= -\frac{2t}{m} \left(1+\frac{t^2}{m}\right)^{-2} \\ \frac{t^2}{m} &= \frac{1}{u} - 1 & & & &= -\frac{2}{u} \sqrt{\frac{m}{u} - m} \cdot u^2 \end{aligned}$$

$$\int_0^{\infty} t^{2R} \left(1+\frac{t^2}{m}\right)^{-\frac{m+1}{2}} dt$$

$$= \int_1^0 \left(\frac{m}{u} - m\right)^R \cdot u^{\frac{m+1}{2}} \cdot \frac{m}{-2 \sqrt{\frac{m}{u} - m} \cdot u^2} du$$

$$= \int_0^1 \frac{1}{2} \left(\frac{m}{u} - m\right)^{R-\frac{1}{2}} \cdot u^{\frac{m-3}{2} - m} du$$

$$= \frac{1}{2} \cdot m^{R+\frac{1}{2}} \int_0^1 (1-u)^{R-\frac{1}{2}} \cdot u^{\frac{m-3}{2} - R + \frac{1}{2}} du$$

$$= \frac{1}{2} \cdot m^{R+\frac{1}{2}} \int_0^1 u^{\left(\frac{m}{2}-R\right)-1} \cdot (1-u)^{\left(R+\frac{1}{2}\right)-1} du$$

$$= \frac{1}{2} \cdot m^{k+\frac{1}{2}} B\left(\frac{m}{2}-k, k+\frac{1}{2}\right)$$

$$= \frac{1}{2} \cdot m^{k+\frac{1}{2}} \cdot \frac{\Gamma(\frac{m}{2}-k) \Gamma(k+\frac{1}{2})}{\Gamma(\frac{m+1}{2})}$$

$$E[T^r] = 2 \cdot \frac{\Gamma(\frac{m+1}{2})}{\Gamma(\frac{m}{2})} \cdot \frac{1}{\sqrt{\pi m}} \cdot \frac{1}{2} m^{k+\frac{1}{2}} \cdot \frac{\Gamma(\frac{m}{2}-k) \Gamma(k+\frac{1}{2})}{\Gamma(\frac{m+1}{2})}$$

$$= \frac{\Gamma(\frac{m}{2}-k) \Gamma(k+\frac{1}{2})}{\Gamma(\frac{m}{2})} \cdot \frac{m^k}{\sqrt{\pi}}$$

$$\underline{0 \leq 2k < m} \quad //$$

$$\underline{\hspace{10em}} //$$

問4

$$Y \sim F_{m,n}$$

$E[Y^r]$ を求めよ.

$$S \sim \chi_m^2, T \sim \chi_n^2$$

$$Y = \frac{S/m}{T/n}$$

$E[Y^r]$

$$= \int_0^\infty y^r \frac{\frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})} \cdot \left(\frac{m}{n}\right)^{\frac{m}{2}} y^{\frac{m}{2}-1}}{\left(1+\frac{m}{n}y\right)^{\frac{m+n}{2}}} dy$$

$$= \frac{\Gamma(\frac{m+n}{2})}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})} \cdot \int_0^\infty \frac{\left(\frac{m}{n}\right)^{\frac{m}{2}} y^{\frac{m}{2}+r-1}}{\left(1+\frac{m}{n}y\right)^{\frac{m+n}{2}}} dy$$

$$\left(1+\frac{m}{n}y\right)^{-1} = t$$

$$\frac{dt}{dy} = -\left(1+\frac{m}{n}y\right)^{-2} \cdot \frac{m}{n}$$

$$= -t^2 \cdot \frac{m}{n}$$

$$dy = -t^{-2} \cdot \frac{n}{m} dt$$

$$\int_0^\infty \left(\frac{m}{n}\right)^{\frac{m}{2}} \cdot \left(1+\frac{m}{n}y\right)^{-\frac{m+n}{2}} \cdot y^{\frac{m}{2}+r-1} dy$$

$$= \int_1^0 \left(\frac{m}{n}\right)^{\frac{m}{2}} t^{\frac{m}{2}} \cdot \left(\frac{n}{m} \cdot \frac{1-t}{t}\right)^{\frac{m}{2}+r-1} \cdot -t^{-2} \cdot \frac{n}{m} \cdot dt$$

$$= \int_0^1 \left(\frac{m}{n}\right)^{-r} t^{\frac{n}{2}-r-1} (1-t)^{\frac{m}{2}+r-1} dt$$

$$= \left(\frac{n}{m}\right)^r \int_0^1 t^{\left(\frac{n}{2}-r\right)-1} \cdot (1-t)^{\left(\frac{m}{2}+r\right)-1} dt$$

$$= \left(\frac{n}{m}\right)^r B\left(\frac{n}{2}-r, \frac{m}{2}+r\right)$$

$$= \left(\frac{n}{m}\right)^r \frac{\Gamma\left(\frac{n}{2}-r\right)\Gamma\left(\frac{m}{2}+r\right)}{\Gamma\left(\frac{m+n}{2}\right)}$$

$E[Y^r]$

$$= \frac{\Gamma\left(\frac{m+n}{2}\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} \cdot \left(\frac{n}{m}\right)^r \frac{\Gamma\left(\frac{n}{2}-r\right)\Gamma\left(\frac{m}{2}+r\right)}{\Gamma\left(\frac{m+n}{2}\right)}$$

$$= \frac{\Gamma\left(\frac{n}{2}-r\right)\Gamma\left(\frac{m}{2}+r\right)}{\Gamma\left(\frac{m}{2}\right)\Gamma\left(\frac{n}{2}\right)} \left(\frac{n}{m}\right)^r$$

$$0 \leq r < \frac{n}{2}$$

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問5

$$X, Y, \text{i.i.d.} \sim N(0, 1)$$

$$Z = \min\{X, Y\} \text{ の } \tau$$

$$Z^2 \sim \chi_1^2 \text{ を示せ.}$$

ZがZより上を  
とためには  
XもYもZより上

$$\begin{aligned} P(Z \leq z) &= P(Z < z) \\ &= 1 - P(z \leq Z) \\ &= 1 - P(z \leq X) \cdot P(z \leq Y) \\ &= 1 - (1 - F_X(z))^2 \end{aligned}$$

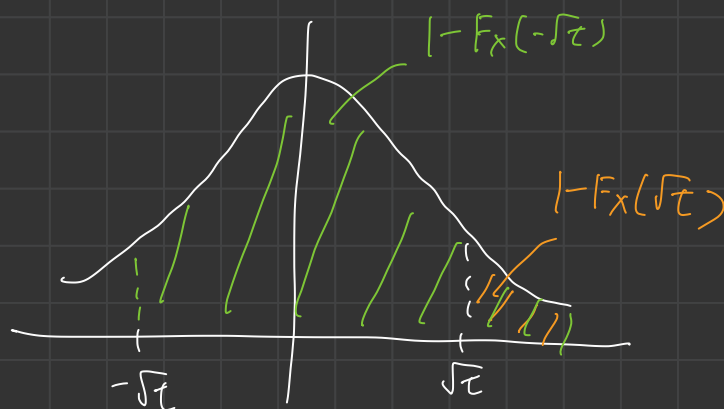
$$f_Z(z) = 2 f_X(z) \cdot (1 - F_X(z))$$

$$T = Z^2$$

$$\begin{aligned} P(T \leq t) &= P(Z^2 \leq t) \\ &= P(-\sqrt{t} \leq Z \leq \sqrt{t}) \end{aligned}$$

$$f_T(t) = \frac{1}{2\sqrt{t}} (f_Z(\sqrt{t}) + f_Z(-\sqrt{t}))$$

$$\begin{aligned} f_T(t) &= \frac{1}{\sqrt{t}} \left( f_X(\sqrt{t}) (1 - F_X(\sqrt{t})) \right. \\ &\quad \left. + f_X(-\sqrt{t}) (1 - F_X(-\sqrt{t})) \right) \\ &= \frac{1}{\sqrt{t}} f_X(\sqrt{t}) \end{aligned}$$



$$f_T(t) = \frac{1}{\sqrt{t}} \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{t}{2}\right\}$$

$$\chi_1^2 = \frac{1}{P(\frac{1}{2})} \left(\frac{1}{2}\right)^{\frac{1}{2}} x^{-\frac{1}{2}} \exp\left\{-\frac{x}{2}\right\}$$

$$= \frac{1}{\sqrt{x}} \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x}{2}\right\}$$

$$\text{よって } T = Z^2 \sim \chi_1^2$$

$$\binom{6}{2} = 15$$

(1)  $Z \sim N(0, 1)$  且  $Z$  服从标准正态分布

$$P(|Z| \geq t) \leq \frac{2}{t} \phi(t) \text{ が成り立つ.}$$

2177 の不等式

$$P(Y \geq c) \leq \frac{E[Y]}{c}$$

$$\begin{aligned} P(|Z| \geq t) &= E[1_{(|Z| \geq t)}] \\ &= \frac{1}{t} E[t 1_{(|Z| \geq t)}] \\ &\leq \frac{1}{t} E[|Z| 1_{(|Z| \geq t)}] \\ &= \frac{1}{t} \left( 2 \cdot \int_t^\infty z \cdot \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} dz \right) \\ &= \frac{2}{t} \left[ -\frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\} \right]_t^\infty \\ &= \frac{2}{t} \left( \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{t^2}{2}\right\} \right) \\ &= \frac{2}{t} \phi(t) \end{aligned}$$

(2) 任意の  $X$  と  $t > 0$

$$P(X \geq a) \leq e^{-at} M_X(t)$$

$$P(X \geq a) = E[1(X \geq a)]$$

$$= e^{-at} E[e^{at} \mathbb{1}(X \geq a)]$$

$$\leq e^{-\lambda t} \mathbb{E}[e^{X_t} \mathbb{1}(X \geq a)]$$

$$\leq e^{-\alpha t} E \left[ e^{X_t} \left( \mathbb{1}(X \geq a) + \mathbb{1}(X < a) \right) \right]$$

$$= e^{-\alpha t} E[e^{xt}]$$

$$= e^{-\alpha t} M_{\kappa}(t)$$



問7

$$g(x) > 0, \quad g(-x) = g(x)$$

$x > 0$  で増加関数

$t > 0$  に對し

$$P(|X| \geq t) \leq \frac{E[g(X)]}{g(t)}$$

$$\begin{aligned} P(|X| \geq t) &= E[\mathbb{1}(|X| \geq t)] \\ &= \frac{1}{g(t)} E[g(t) \mathbb{1}(|X| \geq t)] \end{aligned}$$

$$\leq \frac{1}{g(t)} E[g(X) \mathbb{1}(|X| \geq t)]$$

$$= \frac{1}{g(t)} E\left[g(X) \begin{pmatrix} \mathbb{1}(X \geq t) \\ + \mathbb{1}(X \leq -t) \end{pmatrix}\right]$$

$$\leq \frac{E[g(X)]}{g(t)} \quad (\because g(X) > 0)$$



Fr 8

$$X_1, \dots, X_n, \text{i.i.d.} \sim (\mu, \sigma^2)$$

$$\frac{2}{n(n+1)} \sum_{j=1}^n j X_j \xrightarrow{p} \mu$$

$$E \left[ \left( \frac{2}{n(n+1)} \sum_{j=1}^n j X_j - \mu \right)^2 \right]$$

$$\frac{2}{n(n+1)} = \frac{1}{k}, \quad \sum_{j=1}^n j = \frac{1}{\frac{1}{k}}$$

$$E \left[ \left( \frac{1}{k} \sum_{j=1}^n j X_j - \mu \right)^2 \right]$$

$$= \frac{1}{k^2} E \left[ \left( \sum_{j=1}^n j X_j - \frac{\mu^2}{k} \right)^2 \right]$$

$$= \frac{1}{k^2} E \left[ \left( \sum_{j=1}^n j X_j - \sum_{j=1}^n j \mu \right)^2 \right]$$

$$= \frac{1}{k^2} E \left[ \left( \sum_{j=1}^n j (X_j - \mu) \right)^2 \right]$$

$$= \frac{1}{k^2} E \left[ \sum_{j=1}^n j^2 (X_j - \mu)^2 \right]$$

$$= \frac{1}{k^2} \sum_{j=1}^n j^2 E \left[ (X_j - \mu)^2 \right]$$

$$= \frac{1}{k^2} \sum_{j=1}^n j^2 \sigma^2$$

$$= \frac{2}{n(n+1)} \cdot \frac{2}{n(n+1)} \cdot \frac{n(n+1)(2n+1)}{6} \sigma^2$$

$$= \frac{2(2n+1)}{3n(n+1)} \sigma^2 \rightarrow 0$$

問9

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$V_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

$$(1) \bar{X}_{n+1} = (X_{n+1} + n\bar{X}_n) / (n+1)$$

証明

$$\bar{X}_{n+1} = \frac{1}{n+1} \sum_{i=1}^{n+1} X_i$$

$$= \frac{1}{n+1} \left( X_{n+1} + \sum_{i=1}^n X_i \right)$$

$$= \frac{1}{n+1} (X_{n+1} + n\bar{X}_n)$$

$$= (X_{n+1} + n\bar{X}_n) / (n+1)$$

(2)

$$nV_{n+1}^2 = (n-1)V_n^2 + \frac{n}{n+1} (X_{n+1} - \bar{X}_n)^2$$

証明

$$nV_{n+1}^2 = \sum_{i=1}^{n+1} (X_i - \bar{X}_{n+1})^2$$

$$= \sum_{i=1}^{n+1} \left( X_i - \left( \frac{X_{n+1}}{n+1} + \frac{n}{n+1} \bar{X}_n \right) \right)^2$$

$$= \sum_{i=1}^{n+1} \left( (X_i - \bar{X}_n) - \left( \frac{X_{n+1} - \bar{X}_n}{n+1} \right) \right)^2$$

$$= \sum_{i=1}^{n+1} (X_i - \bar{X}_n)^2 - 2 \sum_{i=1}^{n+1} (X_i - \bar{X}_n) \cdot \frac{X_{n+1} - \bar{X}_n}{n+1}$$

$$+ \sum_{i=1}^{n+1} \left( \frac{X_{n+1} - \bar{X}_n}{n+1} \right)^2$$

$$= (X_{n+1} - \bar{X}_n)^2 + (n-1)V_n^2 - 2 \cdot \frac{1}{n+1} (X_{n+1} - \bar{X}_n)^2 + \frac{(X_{n+1} - \bar{X}_n)^2}{n+1}$$

$$= (n-1)V_n^2 + \frac{n}{n+1} (X_{n+1} - \bar{X}_n)^2$$

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例 10

$X_1, \dots, X_n$, i.i.d., $\sim (\mu, \sigma^2)$

$$\mu_4 = E[(X_i - \mu)^4] \text{ など}$$

$$V_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

(1)

$$\begin{aligned} V_n^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \\ &= \frac{1}{n-1} \sum_{i=1}^n ((X_i - \mu) - (\bar{X}_n - \mu))^2 \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X}_n - \mu) \sum_{i=1}^n (X_i - \mu) + \sum_{i=1}^n (\bar{X}_n - \mu)^2 \right) \end{aligned}$$

$$= \frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X}_n - \mu)(n\bar{X}_n - n\mu) + n(\bar{X}_n - \mu)^2 \right)$$

$$= \frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X}_n - \mu)^2 \right)$$

$$= \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X}_n - \mu)^2 \right)$$

∴

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 = 1 \text{ など}$$

$$Y_i = (X_i - \mu)^2 \text{ など}$$

$$E[Y_i] = E[(X_i - \mu)^2] = \text{var}(X_i) = \sigma^2$$

$$\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 = \bar{Y}_n$$

$$\begin{aligned} \text{var}(Y_i) &= E[(X_i - \mu)^4] \\ &\quad - E[(X_i - \mu)^2]^2 \\ &= \mu_4 - \sigma^4 \end{aligned}$$

よ、

$$Y_i \sim (\sigma^2, \mu_4 - \sigma^4)$$

$$\bar{Y}_n \sim \left(\sigma^2, \frac{\mu_4 - \sigma^4}{n} \right)$$

大数の弱法則より

$$\bar{Y}_n \rightarrow \sigma^2$$

つまり

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 = \sigma^2$$

また

$$\lim_{n \rightarrow \infty} (\bar{X}_n - \mu)^2 = 0$$

大数の弱法則より

$$\bar{X}_n \rightarrow \mu$$

よ、

$$\lim_{n \rightarrow \infty} (\bar{X}_n - \mu)^2 = 0$$

よ、よ、 $\lim_{n \rightarrow \infty} \frac{n}{n-1} = 1$ などより

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X}_n - \mu)^2 \right) \\ = \sigma^2 \end{aligned}$$

(2)

$$X_1, X_2, \dots, X_n \sim (\mu, \sigma^2)$$

$$\sqrt{n}(\bar{X}_n - \mu)$$

$$E[(X_i - \mu)^4] = \mu_4$$

$$= \sqrt{n} \left(\frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X}_n - \mu)^2 \right) - \sigma^2 \right)$$

$$= \sqrt{n} \left(\frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 \right) - \sigma^2 \right)$$

$$= \sqrt{n} \cdot \frac{n}{n-1} \left(\bar{Y}_n - \sigma^2 \right) + \underbrace{\sqrt{n} \left(\frac{n}{n-1} - 1 \right) \sigma^2}_{\rightarrow p 0}$$

$$\sqrt{n}(\bar{X}_n - \mu)^2$$

$$= \sqrt{n}(\bar{X}_n - \mu) \cdot (\bar{X}_n - \mu)$$

$$\rightarrow_d N(0, \sigma^2) \cdot 0 = 0$$

$$\sqrt{n} \cdot \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

以上より

$$\rightarrow_d N(0, \mu_4 - \sigma^4)$$

$$E[(X_i - \mu)^2] = \sigma^2$$

$$E[(X_i - \mu)^4] = \mu_4$$

$$Y_i = (X_i - \mu)^2$$

$$\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

$$Y_i \sim (\sigma^2, \mu_4 - \sigma^4)$$

中心

$$\lim_{n \rightarrow \infty} \sqrt{n}(\bar{Y}_n - \sigma^2) \rightarrow_d N(0, \mu_4 - \sigma^4)$$

問11

$X_1, \dots, X_n$ は (μ, σ^2) の母集団
の標本

$$\text{Cov}(X_i, X_j) = \rho_n \sigma^2, i \neq j$$

$$E[X_i X_j] - E[X_i] E[X_j] = \rho_n \sigma^2$$

$\lim_{n \rightarrow \infty} \rho_n = 0$ のとき $\bar{X}_n = \sum_{i=1}^n \frac{X_i}{n}$ は μ に確率収束

$$\lim_{n \rightarrow \infty} E[(\bar{X}_n - \mu)^2] = 0 \text{ を示す.}$$

$$E\left[\left(\frac{1}{n} \sum_{i=1}^n X_i - \mu\right)^2\right]$$

$$= \frac{1}{n^2} E\left[\left(\sum_{i=1}^n X_i - n\mu\right)^2\right]$$

$$= \frac{1}{n^2} E\left[\left(\sum_{i=1}^n (X_i - \mu)\right)^2\right]$$

$$= \frac{1}{n^2} E\left[\sum_{i=1}^n (X_i - \mu)^2\right]$$

$$+ \frac{1}{n^2} E\left[\sum_{i \neq j} (X_i - \mu)(X_j - \mu)\right]$$

$$= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) + \frac{1}{n^2} \sum_{i \neq j} \rho_n \sigma^2$$

$$= \frac{\text{Var}(X_i)}{n} + \frac{1}{n^2} \cdot n(n-1) \cdot \rho_n \sigma^2$$

$$= \frac{\text{Var}(X_i)}{n} + \frac{n-1}{n} \rho_n \sigma^2$$

$$\lim_{n \rightarrow \infty} \frac{\text{Var}(X_i)}{n} = 0 \quad \lim_{n \rightarrow \infty} \frac{n-1}{n} = 1$$

$$\lim_{n \rightarrow \infty} \rho_n = 0$$

よ $\bar{X}_n$ は μ に平均2乗収束

よ $\bar{X}_n$ は μ に確率収束

問12

$$X_n \sim \text{Bin}(n, p)$$

$$X_n = 0 \text{ のとき } Y_n = 1$$

$$X_n \geq 1 \text{ のとき } Y_n = \log\left(\frac{X_n}{n}\right)$$

このとき

$$Y_n \rightarrow_p \log p$$

$$\sqrt{n}(Y_n - \log p) \rightarrow_d N(0, \frac{1-p}{p})$$

$$\lim_{n \rightarrow \infty} P(|Y_n - \log p| \geq \varepsilon) = 0$$

を示す

$$P(|Y_n - \log p| \geq \varepsilon)$$

$$= P(|Y_n - \log p| \geq \varepsilon, X_n = 0)$$

$$+ P(|Y_n - \log p| \geq \varepsilon, X_n \geq 1)$$

$$= P(|1 - \log p| \geq \varepsilon, X_n = 0)$$

$$+ P\left|\log \frac{X_n}{n} - \log p\right| \geq \varepsilon, X_n \geq 1$$

ここで

$$\lim_{n \rightarrow \infty} P(|1 - \log p| \geq \varepsilon, X_n = 0)$$

$$= \lim_{n \rightarrow \infty} P(|1 - \log p| \geq \varepsilon) P(X_n = 0)$$

$$= P(|1 - \log p| \geq \varepsilon) \lim_{n \rightarrow \infty} P(X_n = 0)$$

$$= P(|1 - \log p| \geq \varepsilon) \lim_{n \rightarrow \infty} (1-p)^n$$

$$= 0$$

$$\lim_{n \rightarrow \infty} P\left|\log \frac{X_n}{n} - \log p\right| \geq \varepsilon$$

$$= \lim_{n \rightarrow \infty} P\left|\log \frac{X_n}{np} \right| \geq \varepsilon$$

$$= \lim_{n \rightarrow \infty} P\left(\log \frac{X_n}{np} \leq -\varepsilon, \varepsilon \leq \log \frac{X_n}{np}\right)$$

$$= \lim_{n \rightarrow \infty} P\left(e^{-\varepsilon} \geq \frac{X_n}{np}, e^{\varepsilon} \leq \frac{X_n}{np}\right)$$

$$= \lim_{n \rightarrow \infty} P\left(pe^{-\varepsilon} \geq \frac{X_n}{n}, pe^{\varepsilon} \leq \frac{X_n}{n}\right)$$

$$= \lim_{n \rightarrow \infty} P\left(pe^{-\varepsilon} - p \geq \frac{X_n - np}{n}, pe^{\varepsilon} - p \leq \frac{X_n - np}{n}\right)$$

$$np \neq -\frac{X_n - np}{n}, \lim_{n \rightarrow \infty} \frac{X_n - np}{n} = 0$$

$$= P(pe^{-\varepsilon} - p \geq 0, pe^{\varepsilon} - p \leq 0)$$

$$= P(e^{-\varepsilon} - 1) < 0, p(e^{\varepsilon} - 1) > 0$$

より この確率は 0

よって

$$\lim_{n \rightarrow \infty} P(|Y_n - \log p| \geq \varepsilon) = 0$$

すなわち

$$Y_n \rightarrow_p \log p$$

□

$$\sqrt{n} (Y_n - \log p) \rightarrow_d N(0, \frac{1-p}{p})$$

$$\sqrt{n} (Y_n - \log p)$$

$$= \sqrt{n} (\log \frac{X_n}{n} - \log p) \mathbb{I}(X_n \geq 1) + \sqrt{n} (1 - \log p) \mathbb{I}(X_n = 0)$$

$$\rightarrow \sqrt{n} (\log \frac{X_n}{n} - \log p)$$

$$\because P(X_n = 0) \xrightarrow[n \rightarrow \infty]{} 0$$

$\sim (1-p)^n$

今

$X_n \sim \text{Bin}(n, p)$ より 中心極限定理から

$$\sqrt{n} (\frac{X_n}{n} - p) \rightarrow_d N(0, p(1-p))$$

$$\left(\begin{array}{l} \frac{X_n}{n} \text{ は } x_1, x_2, \dots, \text{i.i.d.} \sim \text{Ber}(p) \\ \text{としたときの } \bar{X} \text{ と同じ} \end{array} \right)$$

デルタ法より

$$\sqrt{n} (\log \frac{X_n}{n} - \log p) \rightarrow_d \frac{1}{p} N(0, p(1-p))$$

$$= N(0, \frac{1-p}{p})$$

問13

$$\lambda e^{-\lambda x}, x \geq 0$$

$$X_1, \dots, X_n, \text{i.i.d.} \sim \text{Ex}(\lambda)$$

(1) $X_{(1)}$ と $X_{(n)}$ の同時確率密度関数,

$$\begin{aligned} f_{X_{(1)}, X_{(n)}}(x, y) &= \frac{n!}{(n-2)!} f(x) f(y) (F(y) - F(x))^{n-2} \\ &= n(n-1) \lambda^2 e^{-\lambda(x+y)} \\ &\quad \left((1 - e^{-\lambda y}) - (1 - e^{-\lambda x}) \right)^{n-2} \\ &= \frac{n(n-1) \lambda^2 e^{-\lambda(x+y)} (e^{-\lambda x} - e^{-\lambda y})^{n-2}}{\quad} \quad \text{//} \end{aligned}$$

$$\begin{aligned} f_{X_{(j)}}(x) &= \frac{n!}{(j-1)!(n-j)!} \lambda e^{-\lambda x} \\ &\quad \cdot (1 - e^{-\lambda x})^{j-1} (e^{-\lambda x})^{n-j} \\ &= \frac{n!}{(j-1)!(n-j)!} \lambda e^{-\lambda(n-j+1)x} \\ &\quad (1 - e^{-\lambda x})^{j-1} \quad \text{//} \end{aligned}$$

(2)

$U_n = (\lambda X_{(n)} - \log n)$ の漸近分布

$$\begin{aligned} P(U_n \leq x) &= P(\lambda X_{(n)} - \log n \leq x) \\ &= P\left(X_{(n)} \leq \frac{x + \log n}{\lambda}\right) \\ &= P\left(Z_{(n)} \leq \frac{x + \log n}{\lambda}\right) \end{aligned}$$

$$\begin{aligned} \Theta(x) &= \lim_{n \rightarrow \infty} \left\{ F\left(\frac{x + \log n}{\lambda}\right) \right\}^n \\ &= \lim_{n \rightarrow \infty} \left(\int_0^{\frac{x + \log n}{\lambda}} \lambda e^{-\lambda t} dt \right)^n \\ &= \lim_{n \rightarrow \infty} \left(\left[-e^{-\lambda t} \right]_0^{\frac{x + \log n}{\lambda}} \right)^n \\ &= \lim_{n \rightarrow \infty} \left(-e^{-(x + \log n)} + 1 \right)^n \\ &= \lim_{n \rightarrow \infty} \left(-\frac{e^{-x}}{n} + 1 \right)^n \\ &= e^{-e^{-x}} \quad \text{//} \end{aligned}$$

問 14

$$X_1, \dots, X_n, \text{i.i.d.} \sim \text{Ex}(\lambda)$$

$$Y_1 = n X_{(1)}, \quad Y_2 = (n-1)(X_{(2)} - X_{(1)})$$

$$Y_3 = (n-2)(X_{(3)} - X_{(2)})$$

$$Y_n = X_{(n)} - X_{(n-1)}$$

⇔

$$Y_1, \dots, Y_n, \text{i.i.d.} \sim \text{Ex}(\lambda) \text{ を示す.}$$

$$f_{Y_1, Y_2, \dots, Y_n}(y_1, \dots, y_n)$$

$$= f_{X_{(1)}, \dots, X_{(n)}}\left(\frac{y_1}{n}, \frac{y_1}{n} + \frac{y_2}{n-1}, \dots\right) |J|$$

⇔

$$X_{(1)} = \frac{y_1}{n}$$

$$X_{(2)} = \frac{y_1}{n} + \frac{y_2}{n-1}$$

$$X_{(3)} = \frac{y_1}{n} + \frac{y_2}{n-1} + \frac{y_3}{n-2}$$

$$X_{(n)} = \frac{y_1}{n} + \frac{y_2}{n-1} + \frac{y_3}{n-2} + \dots + \frac{y_{n-1}}{2} + \frac{y_n}{1}$$

∴

$$J(Y_1, Y_2, \dots, Y_n) \rightarrow (X_{(1)}, \dots, X_{(n)})$$

$$= \det \begin{pmatrix} \frac{1}{n} & 0 & \dots & \dots & 0 \\ \frac{1}{n} & \frac{1}{n-1} & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{1}{n} & \frac{1}{n-1} & \dots & \frac{1}{2} & 0 \\ \frac{1}{n} & \frac{1}{n-1} & \dots & \frac{1}{2} & 1 \end{pmatrix}$$

$$= \frac{1}{n} \cdot \frac{1}{n-1} \cdot \dots \cdot \frac{1}{1}$$

$$= \frac{1}{n!}$$

∴

$$f_{Y_1, Y_2, \dots, Y_n}(y_1, \dots, y_n)$$

$$= n! \cdot f_{X_{(1)}}\left(\frac{y_1}{n}\right) f_{X_{(2)}}\left(\frac{y_1}{n} + \frac{y_2}{n-1}\right)$$

$$\cdot \dots \cdot f_{X_{(n)}}\left(\frac{y_1}{n} + \dots + \frac{y_n}{1}\right)$$

$$\cdot \frac{1}{n!}$$

$$= f_{X_{(1)}}\left(\frac{y_1}{n}\right) f_{X_{(2)}}\left(\frac{y_1}{n} + \frac{y_2}{n-1}\right)$$

$$\cdot \dots \cdot f_{X_{(n)}}\left(\frac{y_1}{n} + \dots + \frac{y_n}{1}\right)$$

$$= \lambda e^{-\lambda \cdot \frac{y_1}{n}} \cdot \lambda e^{-\lambda \left(\frac{y_1}{n} + \frac{y_2}{n-1}\right)}$$

$$\cdot \dots \cdot \lambda e^{-\lambda \left(\frac{y_1}{n} + \frac{y_2}{n-1} + \dots + \frac{y_n}{1}\right)}$$

$$= \lambda e^{-\lambda \cdot y_1} \cdot \lambda e^{-\lambda y_2}$$

$$\cdot \dots \cdot \lambda e^{-\lambda y_n}$$

∴

$$Y_i \sim \lambda e^{-\lambda y_i} = \text{Ex}(\lambda)$$

∴ 独立同分布.

問15

$X_1, \dots, X_n$ は互いに独立, 非負

X_i のハザード関数 $\lambda_i(x)$

(1) $X_{(n)}$ のハザード関数は

$$\lambda_{X_{(n)}}(x) = \lambda_1(x) + \dots + \lambda_n(x)$$

$$\lambda_i(x) = \frac{f_{X_i}(x)}{1 - F_{X_i}(x)} \quad \begin{array}{l} X_{(n)} \geq x \text{ とき} \\ X_1, \dots, X_n \geq x \end{array}$$

$$\begin{aligned} \lambda_{X_{(n)}}(x) &= \frac{f_{X_{(n)}}(x)}{1 - F_{X_{(n)}}(x)} \\ &= \frac{d}{dx} \underbrace{-\log(1 - F_{X_{(n)}}(x))} \\ &= \frac{d}{dx} \underbrace{-\log \prod_{i=1}^n (1 - F_{X_i}(x))} \\ &= \frac{d}{dx} \left(\begin{array}{l} -\log(1 - F_{X_1}(x)) \\ -\log(1 - F_{X_2}(x)) \\ \dots \\ -\log(1 - F_{X_n}(x)) \end{array} \right) \\ &= \frac{f_{X_1}(x)}{1 - F_{X_1}(x)} + \dots + \frac{f_{X_n}(x)}{1 - F_{X_n}(x)} \\ &= \lambda_1(x) + \dots + \lambda_n(x) \end{aligned}$$

$$(2) \lambda_{X_{(n)}}(x) \leq \lambda_1(x) + \dots + \lambda_n(x)$$

$$F_{X_{(n)}}(x) = \prod_{i=1}^n F_{X_i}(x)$$

$$\begin{aligned} f_{X_{(n)}}(x) &= \frac{d}{dx} \prod_{i=1}^n F_{X_i}(x) \\ &= \sum_{i=1}^n f_{X_i}(x) \prod_{j \neq i} F_{X_j}(x) \end{aligned}$$

$$\begin{aligned} \lambda_{X_{(n)}} &= \frac{\sum_{i=1}^n f_{X_i}(x) \prod_{j \neq i} F_{X_j}(x)}{1 - \prod_{i=1}^n F_{X_i}(x)} \\ &= \sum_{i=1}^n \left(\frac{f_{X_i}(x) \prod_{j \neq i} F_{X_j}(x)}{1 - \prod_{k=1}^n F_{X_k}(x)} \right) \end{aligned}$$

$$\leq \sum_{i=1}^n \frac{f_{X_i}(x)}{1 - F_{X_i}(x)} = \lambda_1(x) + \dots + \lambda_n(x)$$

この不等式は
まず

$$1 - \prod_{k=1}^n F_{X_k} \leq 1 - F_{X_i}(x)$$

また

$$F_{X_j}(x) \leq 1 \text{ より}$$

$$\prod_{j \neq i} F_{X_j}(x) \leq 1$$

$$\text{よって } f_{X_i}(x) \prod_{j \neq i} F_{X_j}(x) \leq f_{X_i}(x)$$

問16

$X_1, \dots, X_n$, i.i.d. $\sim f(x|\theta)$

$$f(x|\theta) = \frac{a}{\theta^a} x^{a-1} \\ 0 < x < \theta \\ a > 0$$

変換する

$$U_1 = \frac{X_{(1)}}{X_{(2)}}, \quad U_2 = \frac{X_{(2)}}{X_{(3)}}$$

$$U_{n-1} = \frac{X_{(n-1)}}{X_{(n)}}, \quad U_n = X_{(n)}$$

と置く

(1)

$(U_1, \dots, U_n)$ の同時確率密度関数

$$X_{(n)} = U_n, \quad X_{(n-1)} = U_n U_{n-1}$$

$$X_{(n-2)} = U_n U_{n-1} U_{n-2}$$

$$X_1 = U_1 \cdot U_2 \cdot \dots \cdot U_n$$

$$f_{U_1, U_2, \dots, U_n}(u_1, u_2, \dots, u_n)$$

$$= f_{X_{(1)}, X_{(2)}, \dots, X_{(n)}}(U_1 \cdot U_2 \cdot \dots \cdot U_n, \dots, U_{n-1} U_n, U_n) \\ |J|$$

$$J([U_1, U_2, \dots, U_n] \rightarrow [X_{(1)}, \dots, X_{(n)}])$$

$$= \det \begin{pmatrix} 0 & \dots & 0 & U_n & U_{n-1} \\ 0 & \dots & 0 & 1 & 0 \end{pmatrix}$$

$$1 \cdot U_n = (U_n U_{n-1}) (U_n U_{n-1} U_{n-2})$$

$$\dots = (U_n U_{n-1} U_{n-2} \dots U_2)$$

$$= U_2 \cdot U_3^2 \cdot \dots \cdot U_n^{n-1}$$

$$f_{U_1, \dots, U_n}(u_1, \dots, u_n)$$

$$= n! \cdot u_2 \cdot u_3^2 \cdot \dots \cdot u_n^{n-1}$$

$$f(u_1, \dots, u_n) \cdot f(u_2, \dots, u_n) \cdot \dots \cdot f(u_n)$$

$$= n! \cdot u_2 \cdot u_3^2 \cdot \dots \cdot u_n^{n-1}$$

$$\cdot \frac{a}{\theta^a} (u_1 \cdot \dots \cdot u_n)^{a-1} \cdot \dots \cdot \frac{a}{\theta^a} (u_n)^{a-1}$$

$$= \frac{a}{\theta^a} u_1^{a-1} \cdot \frac{2a}{\theta^a} u_2^{2a-1}$$

$$\cdot \dots \cdot \frac{na}{\theta^a} \cdot u_n^{na-1}$$

$$f_{X_{(n)}}(x) = n f(x) \{F(x)\}^{n-1}$$

$$= n \cdot \frac{a}{\theta^a} x^{a-1} \cdot \left(\frac{x}{\theta}\right)^{n-1}$$

$$= na \cdot \frac{x^{na-1}}{\theta^{na}}$$

(2)

$$F(x) = \frac{x^a}{\theta^a}$$

$$F\left(\frac{U_n - b_n}{a_n}\right) = P\left(\frac{U_n - b_n}{a_n} \leq x\right)$$

$$= P(U_n \leq a_n x + b_n)$$

$$= F_{U_n}(a_n x + b_n)$$

$$= \left(\frac{a_n x + b_n}{\theta}\right)^{na}$$

$$b_n = 0, \quad a_n = \frac{\theta}{n}$$

$$\left(1 + \frac{x}{\theta n}\right)^{na}$$

$$\rightarrow (e^x)^a$$

$$= \exp\{a x\}$$

問 17

$$X_1, \dots, X_n, \text{i.i.d.} \sim F(x)$$

$$\alpha > 0$$

$$\lim_{x \rightarrow \infty} x^\alpha (1 - F(x)) = \lambda > 0$$

$$(1) \quad \forall x > 0$$

$$\lim_{n \rightarrow \infty} n [1 - F(n^{\frac{1}{\alpha}} x)]$$

$$= \lambda x^{-\alpha}$$

$$\lim_{x \rightarrow \infty} x^\alpha (1 - F(x)) = \lambda \quad \text{の } x \text{ に } n^{\frac{1}{\alpha}} x \text{ を代入}$$

$$\lim_{n^{\frac{1}{\alpha}} x \rightarrow \infty} (n^{\frac{1}{\alpha}} x)^\alpha (1 - F(n^{\frac{1}{\alpha}} x))$$

$$= \lim_{n^{\frac{1}{\alpha}} x \rightarrow \infty} n \cdot x^\alpha (1 - F(n^{\frac{1}{\alpha}} x)) = \lambda$$

$$n \rightarrow \infty \text{ のとき } n^{\frac{1}{\alpha}} x \rightarrow \infty \text{ かつ } (\because \alpha > 0, x > 0)$$

$$\lim_{n \rightarrow \infty} n \cdot x^\alpha [1 - F(n^{\frac{1}{\alpha}} x)] = \lambda$$

$x^{-\alpha}$ は定数 x に対して変わらない。

$$\lim_{n \rightarrow \infty} n [1 - F(n^{\frac{1}{\alpha}} x)] = \lambda x^{-\alpha}$$

□

(2)

$$Z_n = X_{(n)}$$

$$x > 0 \text{ に対して}$$

$$P(n^{-\frac{1}{\alpha}} Z_n \leq x) \rightarrow \exp\{-\lambda x^{-\alpha}\}$$

を示す。

$$P(n^{-\frac{1}{\alpha}} Z_n \leq x)$$

$$= P(Z_n \leq n^{\frac{1}{\alpha}} x)$$

$$= (F(n^{\frac{1}{\alpha}} x))^n$$

$$= \left(\frac{n - n[1 - F(n^{\frac{1}{\alpha}} x)]}{n} \right)^n$$

$$= \left(1 - \frac{n[1 - F(n^{\frac{1}{\alpha}} x)]}{n} \right)^n$$

$$\rightarrow \exp\{-\lambda x^{-\alpha}\}$$

$$\left(\begin{array}{l} \text{収束の数列 } (c_n) \text{ (} c_n \rightarrow c \text{) に対して} \\ \therefore \lim_{n \rightarrow \infty} \left(1 + \frac{c_n}{n} \right)^n \rightarrow \exp\{c\} \\ \text{が成立.} \end{array} \right)$$

(3)

$$\lim_{n \rightarrow \infty} P(n^{-\frac{1}{\alpha}} Z_n \leq 0) = 0$$

を示す.

$$\lim_{n \rightarrow \infty} P(n^{-\frac{1}{\alpha}} Z_n \leq 0)$$

$$= \lim_{n \rightarrow \infty} P(Z_n \leq 0)$$

$$= (F(0))^n = 0$$

$\therefore F(0) < 1$ を示す.

$F(0) = 1$ と仮定すると

$$\forall x > 0 \text{ で } F(x) = 1.$$

すると

$$\lim_{x \rightarrow \infty} x^\lambda (1 - F(x)) = \lim_{x \rightarrow \infty} 0$$

$$= 0 \neq \lambda \quad (\because \lambda > 0)$$

以上より

$F(0) < 1$ が示せたから

$$\lim_{n \rightarrow \infty} P(n^{-\frac{1}{\alpha}} Z_n \leq 0) = 0 \text{ が示せた.}$$

問 18

$$X_1, \dots, X_n, \text{i.i.d.} \sim F(x)$$

$$X_i \geq 0 \quad M_n = X_{(1)}$$

$$(1) \lim_{x \rightarrow 0} \frac{F(x)}{x} = \lambda > 0 \text{ と仮定.}$$

$$nM_n \rightarrow_d \text{Ex}(\lambda) \text{ と示す.}$$

$$nM_n \rightarrow_d \text{Ex}(\lambda) \text{ とは}$$

$$\lim_{n \rightarrow \infty} P(nM_n \leq x) = 1 - e^{-\lambda x}$$

と示せばよい.

$$\lim_{n \rightarrow \infty} P(nM_n \leq x)$$

$$= \lim_{n \rightarrow \infty} P(M_n \leq \frac{x}{n})$$

$$= \lim_{n \rightarrow \infty} 1 - P(M_n \geq \frac{x}{n})$$

$$= \lim_{n \rightarrow \infty} 1 - (1 - F(\frac{x}{n}))^n$$

$$= \lim_{n \rightarrow \infty} 1 - \left(1 - \frac{F(\frac{x}{n})}{\frac{x}{n}} \cdot x\right)^n$$

$$\because \lim_{n \rightarrow \infty} \frac{F(\frac{x}{n})}{\frac{x}{n}} = \lambda \text{ かつ}$$

$$\lim_{n \rightarrow \infty} 1 - \left(1 - \frac{x}{n} \cdot \lambda\right)^n$$

$$= 1 - e^{-\lambda x}$$

より示せた.

(2)

ある $\alpha > 0$ に対して

$$\lim_{x \rightarrow 0} \frac{F(x)}{x^\alpha} = \lambda > 0 \text{ と仮定.}$$

$n^{\frac{1}{\alpha}} M_n$ は λ の指数分布に収束する

$$\lim_{n \rightarrow \infty} P(n^{\frac{1}{\alpha}} M_n \leq x)$$

$$= \lim_{n \rightarrow \infty} P(M_n \leq \frac{x}{n^{\frac{1}{\alpha}}})$$

$$= \lim_{n \rightarrow \infty} 1 - P(M_n \geq \frac{x}{n^{\frac{1}{\alpha}}})$$

$$= \lim_{n \rightarrow \infty} 1 - \left(1 - F(\frac{x}{n^{\frac{1}{\alpha}}})\right)^n$$

$$= \lim_{n \rightarrow \infty} 1 - \left(1 - \frac{F(\frac{x}{n^{\frac{1}{\alpha}}})}{\frac{x^\alpha}{n}} \cdot x^\alpha\right)^n$$

$$= 1 - \exp\{-\lambda x^\alpha\}$$

$$\lim_{n \rightarrow \infty} \frac{F(\frac{x}{n^{\frac{1}{\alpha}}})}{\frac{x^\alpha}{n}}$$